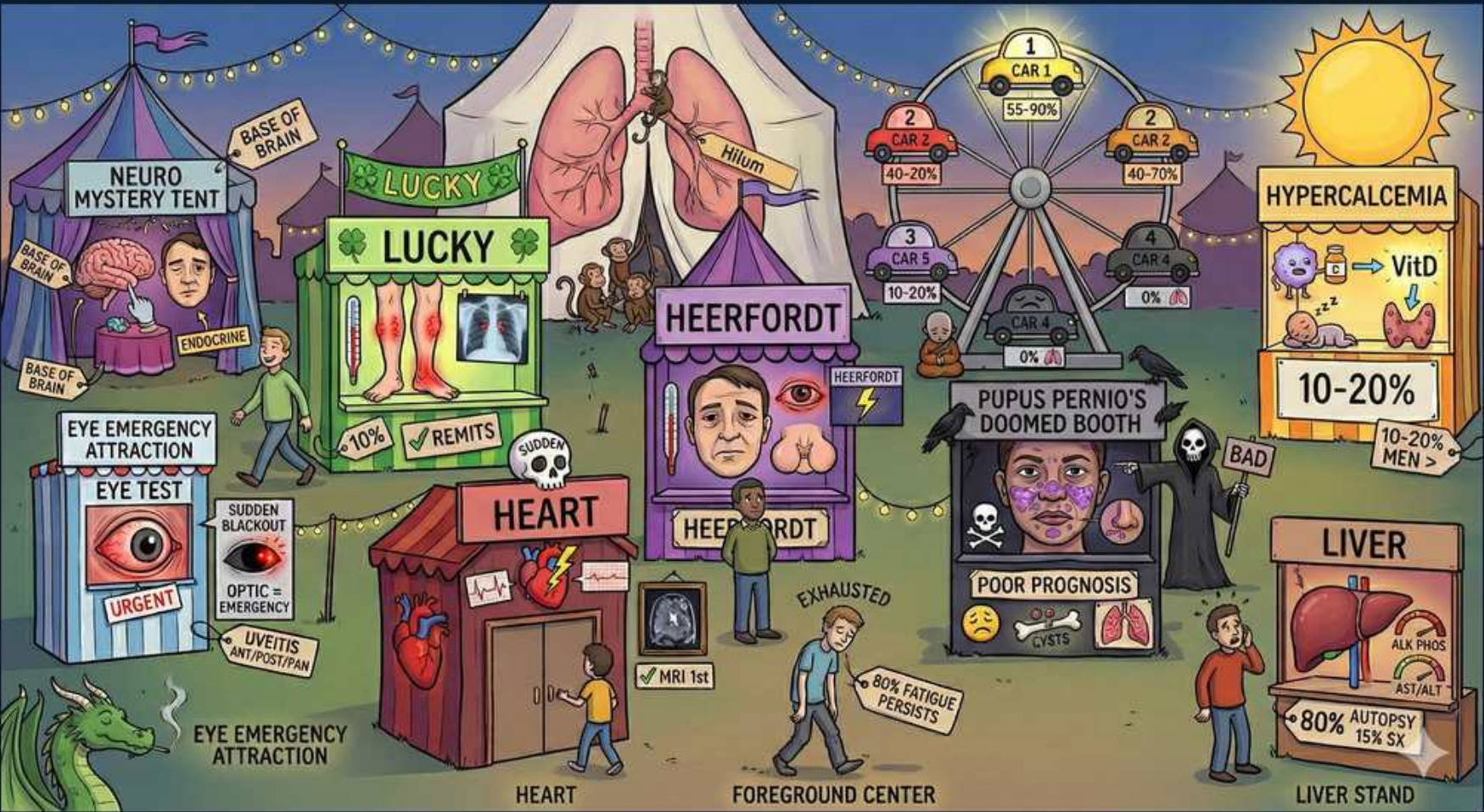


Sarcoidosis — Organ Involvement



1. Neuro (base of brain)

WHAT YOU SEE

A purple 'NEURO MYSTERY TENT' top-left with sad-faced figures and repeated BASE OF BRAIN labels — the mystery/base-of-brain theme = granulomas infiltrating the basilar meninges and cranial nerves.

WHAT IT ENCODES

Neurosarcoidosis affects ~5-10% of patients but is organ-threatening and demands aggressive therapy. Granulomatous infiltration favors the basilar meninges → the cranial nerve VII (facial) palsy is the single most common manifestation (often bilateral, a classic board clue), followed by optic neuritis (CN II). Basilar/hypothalamic involvement causes pituitary dysfunction. Diagnosis: gadolinium-enhanced brain MRI shows leptomeningeal enhancement; CSF shows lymphocytic pleocytosis and high CSF ACE. Treat with high-dose steroids ± infliximab.

BOARD TRAP

Calling isolated bilateral facial palsy 'idiopathic Bell palsy' — bilateral CN VII palsy is a red flag for neurosarcoidosis (and Lyme, GBS), NOT idiopathic Bell palsy. Work it up with MRI/CSF rather than just steroids-and-discharge.

2. Löfgren = LUCKY

WHAT YOU SEE

A bright green carnival booth plastered with four-leaf clovers and the word LUCKY — the lucky clovers = Löfgren's excellent, self-remitting prognosis.

WHAT IT ENCODES

Löfgren syndrome = the classic triad of erythema nodosum + bilateral hilar lymphadenopathy + migratory polyarthritis (often ankles), ± fever. It is the acute, self-limited form with an EXCELLENT prognosis — spontaneous remission in ~80-90% within 1-2 years. Strongly associated with HLA-DRB1*0301. Because the clinical picture is so characteristic, Löfgren is one of the few sarcoid presentations that can be diagnosed WITHOUT a confirmatory biopsy.

BOARD TRAP

Aggressively immunosuppressing Löfgren with chronic steroids — it usually self-resolves, so first-line is just NSAIDs (and colchicine for the arthritis). Reaching for prednisone or methotrexate here is the wrong answer; reserve those for symptomatic or organ-threatening chronic disease.

3. Löfgren remits ~10%? checkmark

WHAT YOU SEE

A green checkmark on the LUCKY booth beside 'REMITTS' — the checkmark = spontaneous remission (the good-prognosis acute phenotype).

WHAT IT ENCODES

Löfgren is the good-prognosis acute phenotype that largely remits spontaneously (~80-90% within 1-2 years). The presence of erythema nodosum in general is a marker of GOOD prognosis in sarcoid, in contrast to lupus pernio which marks chronic/poor-prognosis disease.

BOARD TRAP

Assuming acute, dramatic onset (high fever, painful nodules, swollen ankles) means worse disease — in sarcoid the opposite is true: acute Löfgren-type presentation predicts remission, while insidious/chronic onset predicts persistent fibrotic disease.

4. Eye emergency / uveitis

WHAT YOU SEE

Eye Test attraction, uveitis, urgent

WHAT IT ENCODES

Ocular involvement occurs in ~25-50% of patients; anterior uveitis is the most common form and can be the presenting feature. Symptoms: red painful eye, photophobia, blurred vision; chronic uveitis causes synechiae, glaucoma, and blindness if untreated. All new sarcoid patients need a baseline slit-lamp exam. Treat with topical/systemic steroids; refractory disease gets methotrexate or anti-TNF.

BOARD TRAP

Treating a red eye in sarcoid as simple conjunctivitis — sarcoid causes UVEITIS (ciliary flush, pain, photophobia, hypopyon), not the painless discharge of conjunctivitis. Missing it risks permanent vision loss. Also remember uveitis + parotitis + fever + facial palsy = Heerfordt syndrome.

5. Heerfordt syndrome

WHAT YOU SEE

Heerfordt booth with sad face

WHAT IT ENCODES

Heerfordt syndrome (uveoparotid fever) = anterior uveitis + parotid gland enlargement + facial nerve (CN VII) palsy + fever. It is a pathognomonic constellation that essentially clinches a sarcoid diagnosis when present. The facial palsy results from CN VII compression as it passes through the inflamed parotid.

BOARD TRAP

Attributing parotid swelling + dry eyes/mouth to Sjögren syndrome — Heerfordt's parotid enlargement plus uveitis and facial palsy points to sarcoid; biopsy shows non-caseating granulomas (Sjögren shows lymphocytic infiltrate, positive anti-Ro/La). They can look alike on the surface but the histology and serology differ.

6. Cardiac sarcoid

WHAT YOU SEE

HEART booth with MRI label

WHAT IT ENCODES

Cardiac sarcoid is a leading cause of sarcoid death. Granulomas infiltrate the conduction system and myocardium → high-grade AV block (classic in a young patient), ventricular arrhythmias, dilated/restrictive cardiomyopathy, and sudden cardiac death. Cardiac MRI (late gadolinium enhancement) and FDG-PET are the key tests. Treat with steroids ± immunosuppression; many need an ICD for arrhythmia/sudden-death prevention.

BOARD TRAP

Relying on a normal resting ECG to exclude cardiac sarcoid — a normal ECG does NOT rule it out. Unexplained high-grade AV block or VT in a young patient should trigger cardiac MRI/PET. Also a wrong move: giving only an antiarrhythmic without an ICD when arrhythmic sudden-death risk is present.

7. Lupus pernio

WHAT YOU SEE

Lupus Pernio's Doomed Booth, poor prognosis

WHAT IT ENCODES

Lupus pernio = chronic violaceous, indurated plaques on the nose, cheeks, ears, and lips. It is the most SPECIFIC cutaneous sign of sarcoidosis and marks chronic, treatment-resistant, poor-prognosis disease, often with associated upper-respiratory tract and pulmonary fibrosis. Frequently needs anti-TNF therapy (infliximab) because it responds poorly to steroids alone.

BOARD TRAP

Confusing lupus pernio with systemic lupus erythematosus (SLE) — despite the name, lupus pernio is a SARCOID lesion with NO relation to SLE. It is also distinct from erythema nodosum: lupus pernio = bad prognosis/chronic, erythema nodosum = good prognosis/acute. Mixing those two prognostic implications is a classic trap.

8. Hypercalcemia 10-20%

WHAT YOU SEE

Hypercalcemia booth with VitD, 10-20%

WHAT IT ENCODES

In ~10-20% of patients, activated macrophages within granulomas express 1-alpha-hydroxylase, which converts 25-OH vitamin D to active 1,25-(OH)₂ vitamin D (calcitriol) INDEPENDENT of PTH. This drives gut calcium absorption → hypercalcemia and hypercalciuria (nephrolithiasis, nephrocalcinosis, AKI). Labs: high Ca, high 1,25-vit D, LOW PTH, LOW 25-OH vit D. Treat with steroids (block 1-alpha-hydroxylase), hydroxychloroquine, hydration, and AVOID sunlight/vitamin D/calcium.

BOARD TRAP

Working up sarcoid hypercalcemia as primary hyperparathyroidism — in sarcoid the PTH is SUPPRESSED (low), driven by granuloma-made calcitriol, whereas primary hyperparathyroidism has HIGH PTH. Another trap: 'correcting' the low 25-OH vitamin D with supplements or recommending sun exposure — that worsens the hypercalcemia.

9. Liver (ALK PHOS)

WHAT YOU SEE

Liver stand, ALK PHOS, 80% autopsy 15% sx

WHAT IT ENCODES

Hepatic granulomas are found in up to ~80% at autopsy/biopsy but are clinically symptomatic in only ~15%. The biochemical clue is an isolated, cholestatic-pattern elevated alkaline phosphatase (± GGT). Most cases need no specific therapy; symptomatic cholestasis is treated with steroids or ursodeoxycholic acid.

BOARD TRAP

Assuming an elevated alkaline phosphatase in sarcoid is from bone disease — fractionate it or check GGT to confirm the hepatic source. Also do not anchor on infection: granulomatous hepatitis has a broad differential (TB, fungal, PBC, drugs), so a liver biopsy showing non-caseating granulomas still requires excluding these mimics.

10. Erythema nodosum

WHAT YOU SEE

Tender red shin nodules near booths

WHAT IT ENCODES

Erythema nodosum = tender, erythematous, ill-defined nodules on the anterior shins; it is a reactive panniculitis (septal, no vasculitis) and a component of Löfgren syndrome. In sarcoid it is a GOOD-prognosis marker. It is non-specific overall (also seen with strep, IBD, TB, drugs, pregnancy), so context matters.

BOARD TRAP

Biopsying the erythema nodosum nodule to diagnose sarcoid — EN itself shows only a nonspecific panniculitis, NOT granulomas. Biopsy a more specific site (skin granuloma, lymph node) instead. Treat EN supportively with NSAIDs.

11. Fatigue persists

WHAT YOU SEE

Exhausted figure labeled fatigue persists

WHAT IT ENCODES

Fatigue is one of the most common and disabling symptoms of sarcoidosis and can persist even after measurable organ inflammation has resolved or been treated. Contributors include the inflammatory burden itself, small-fiber neuropathy, sleep disturbance, depression, and steroid side effects.

BOARD TRAP

Escalating immunosuppression solely because of persistent fatigue — fatigue often does NOT track with active granulomatous inflammation, so more steroids/biologics may not help and add toxicity. Evaluate for small-fiber neuropathy, hypothyroidism, depression, and sleep apnea instead.

12. Pulmonary center

WHAT YOU SEE

Large lungs at top center of the fair

WHAT IT ENCODES

The lung is the dominant organ, involved in >90% of patients. Pathophysiology: granulomas in the interstitium and bronchovascular bundles cause a RESTRICTIVE pattern on PFTs (low DLCO, low FVC). Chest staging: Stage 0 = normal; Stage I = bilateral hilar adenopathy alone; Stage II = adenopathy + infiltrates; Stage III = infiltrates without adenopathy; Stage IV = fibrosis. Higher stage = worse prognosis.

BOARD TRAP

Expecting an obstructive PFT pattern — sarcoid is classically RESTRICTIVE with reduced DLCO (although endobronchial disease can cause an obstructive overlay). Also, don't equate Stage I (just adenopathy) with severe disease: Stage I is asymptomatic and observed, while Stage IV fibrosis is the bad one.

13. Endocrine / pituitary

WHAT YOU SEE

An ENDOCRINE label tucked inside the purple neuro tent — placing endocrine disease in the brain tent = hypothalamic-pituitary granulomas causing central DI.

WHAT IT ENCODES

Neurosarcoid favoring the hypothalamic-pituitary axis causes central (cranial) diabetes insipidus — polyuria/polydipsia from low ADH — plus possible anterior pituitary failure (hypogonadism, hypothyroidism, adrenal insufficiency). DI is confirmed by the water-deprivation test and responds to desmopressin.

BOARD TRAP

Mislabeling sarcoid polyuria as nephrogenic DI or osmotic diuresis from hypercalcemia — central DI from pituitary granulomas responds to desmopressin (urine concentrates), whereas nephrogenic DI does not. Note hypercalcemia in sarcoid can independently cause nephrogenic DI, so check the calcium too.

14. Sudden blackout / optic emergency

WHAT YOU SEE

An eye-attraction sign reading SUDDEN BLACKOUT / OPTIC = EMERGENCY — the blacked-out eye = acute optic-nerve sarcoid vision loss demanding urgent IV steroids.

WHAT IT ENCODES

Acute visual loss from optic nerve sarcoidosis (granulomatous optic neuritis) is a sight-threatening ophthalmologic emergency requiring urgent high-dose IV corticosteroids to preserve vision. It is a manifestation of neurosarcoidosis (CN II).

BOARD TRAP

Managing acute monocular vision loss in a sarcoid patient with a routine outpatient referral — optic neuritis here is an emergency: give urgent high-dose steroids and obtain MRI. Delay risks permanent blindness. Don't confuse it with the (also urgent) anterior uveitis — both threaten vision but the optic neuritis is a CNS/neurosarcoid lesion.
